

MAT 310, Handout 6: Probability Distribution, Expected Value, and Variance of Discrete Random Variables (2.11, 3.1-3.3)

Definition: Random Variable A random variable X is a function that assigns a _____ x to each outcome in $\mathbb{S}$.

A **discrete** RV is one which can take on a finite (or countable) number of values.

$\{X = x\}$ is the set of all sample points in $\mathbb{S}$ assigned to the value x by the RV X .

$P(X = x) = p(x)$ = sum of the probabilities of all sample points in $\{X = x\}$.

We know $p(x) \geq$ _____ for all x , and also that $\sum_x p(x) =$ _____

Definition: Probability Distribution of a Discrete RV $\mathbf{X}$ is the set of all values of x and the corresponding probabilities $p(x)$, generally given by a formula, table, or graph.

Note: An alternative name for the *probability distribution* of a discrete RV is *probability mass function* (*pmf*).

Example 1. Roll 2 dice. Find the probability distribution of the sum of the two dice.

Example 2. Draw 2 cards from a deck. Let $X = \#$ of face cards (Jack, Queen, King) drawn. Find the probability distribution of X .

The **expected value** of a discrete RV X is given by:

$$\mu_X = E(X) = \sum_x x \cdot p(x).$$

The expected value of some function of X , say $h(X)$, is given by:

$$E(h(X)) = \sum_x h(x) \cdot p(x).$$

Example 3. Find the expected value of X in Example 2.

Example 4. Consider the RV Y with the distribution given by:

y	$p(y)$
0	.1
1	.2
2	.3
3	.4

Find $E(Y)$ and $E(Y^2)$.

Properties of Expectation

1. $E(c) =$

2. $E[c \cdot g(Y)] =$

3. $E[g_1(Y) + g_2(Y) + \dots + g_k(Y)] =$

Variance of RV Y

$$V(Y) =$$

where $\mu_Y = E(Y)$, and the standard deviation of Y is given by $\sigma_Y = \sqrt{V(Y)} \geq 0$.

Note: subscripts for μ and σ may be omitted when the random variable of interest is clear (e.g., when there is only one random variable).

Important Result: $V(Y) = E(Y^2) - [E(Y)]^2$.

Proof:

Example 5. Find $V(Y)$ for the RV Y in Example 4.

From the results of Exercise 3.33(b) in the HW, we have:

$$V(aY + b) = a^2 V(Y) = a^2 \sigma^2.$$

Example 6. Find $V(2Y + 10)$ for the RV Y in Example 4.